SIMATS ENGINEERING

ASSESSMENT TOOL 3 – Assignment 2

COURSE NAME: STATISTICS WITH R PROGRAMMING
COURSE CODE: ITA04

COURSE OUTCOME COVERED CO3:

Construct and manipulate data frames using reshaping, merging, and file I/O operations in R to prepare structured datasets for analysis. (BL3, BL5)

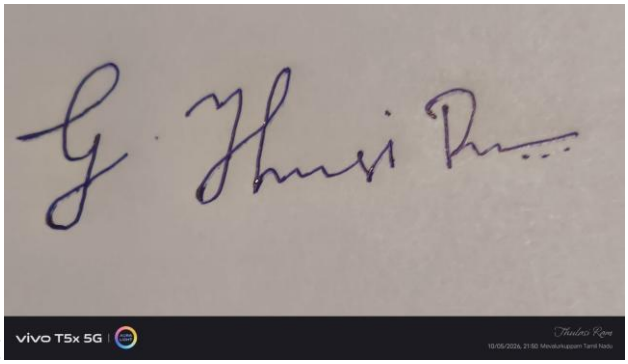
Assessment Tool Weightage: 20%

QUESTIONS: 5 Total Marks: 100

Annexure Assignment 2

Code of Conduct:

I Gunduboina Thulasi Ram(1925260190) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assignment 2:

- Generate 5 Questions for your Capstone Project module
- Apply the formulae and use the relevant graph to visualize it
- Upload the file in Github link
- Save the 5 Questions and Gihub link in the Excell sheet

Question 1: A student wants to analyze monthly expenses across different categories such as Food, Transport, Education, Shopping, and Entertainment. Create a dataset containing expense categories and their corresponding amounts. Display a bar graph or pie chart to compare expenses across categories and identify the category with the highest expenditure.

Formula: **ANOVA F-statistic formula**

$$F = MSB/MSW$$

R Program:

```
category <- c("Food", "Transport", "Education", "Shopping", "Entertainment")

amount <- c(3000, 1500, 4000, 2500, 1000)

expenses <- data.frame(category, amount)

print(expenses)

library(ggplot2)

ggplot(expenses, aes(x = category, y = amount))

 + geom_bar(stat = "identity", fill = "skyblue") +

 labs(title = "Monthly Expenses by Category",

x = "Expense Category",

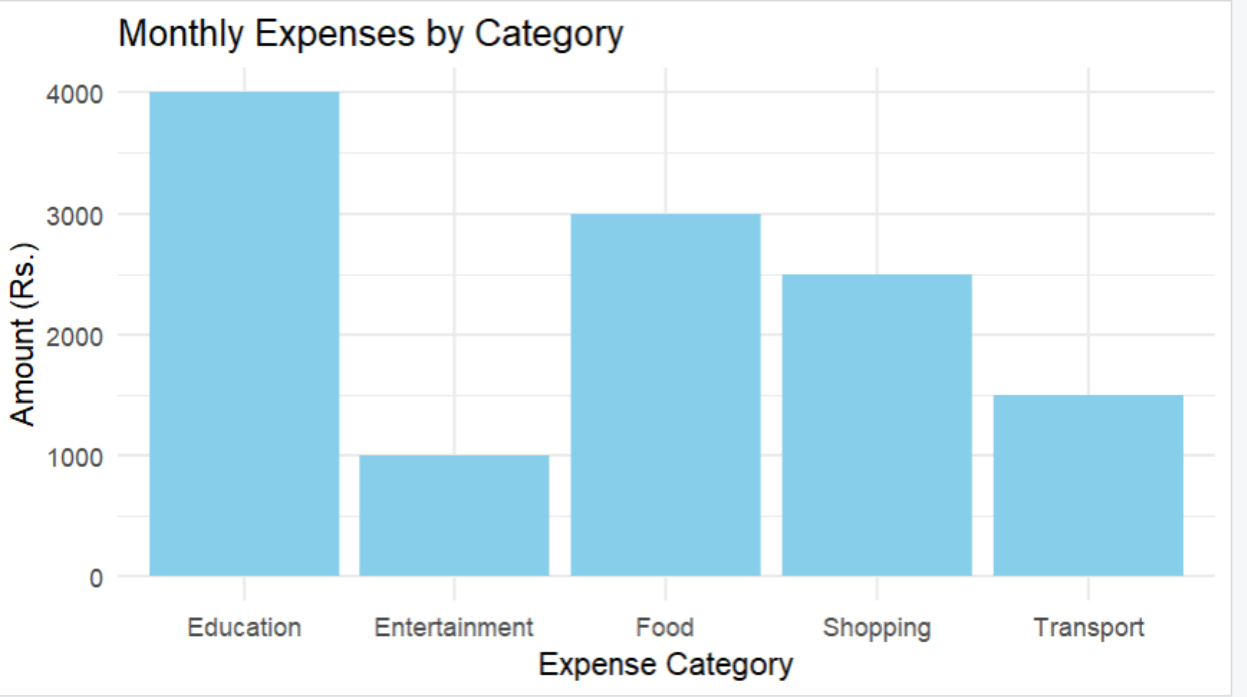
 y = "Amount (Rs.)") +

 theme_minimal()

highest <- expenses[which.max(expenses$amount), ]

print(highest)
```

Output:



Graph: Bar Graph

Question2: A family wants to monitor its monthly budget by comparing the planned budget with actual expenses for six months. Create a dataset containing Month, Budget Amount, and Actual Expense. Display a line graph to compare budgeted and actual expenses and identify months in which spending exceeds the planned budget.

Formula: $$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

R Program:

```
month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun")

budget <- c(10000, 12000, 11000, 13000, 12500, 14000)

actual <- c(9500, 12500, 10500, 14000, 12000, 15000)

expenses <- data.frame(month, budget, actual)

print(expenses)

library(ggplot2)

library(reshape2)

long_data <- melt(expenses, id.vars = "month", variable.name = "Type", value.name =
"Amount")

ggplot(long_data, aes(x = month, y = Amount, color = Type, group = Type)) +

 geom_line(linewidth = 1) +

 geom_point(size = 3) +

 labs(title = "Monthly Budget vs Actual Expenses",

 x = "Month",

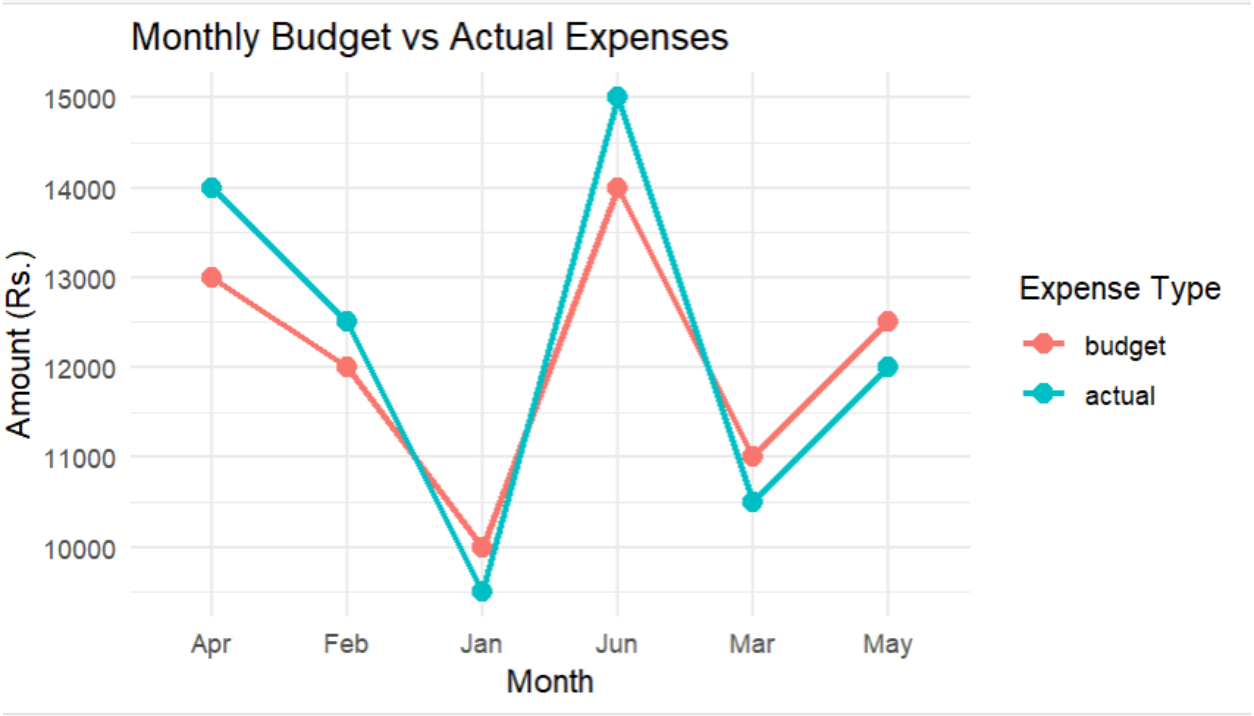
y = "Amount (Rs.)",

 color = "Expense Type") + theme_minimal()

exceeded <- expenses[expenses$actual > expenses$budget, ]

print(exceeded)
```

Output:



Question 3: A student records daily spending on weekdays and weekends. Create a dataset and display a bar chart comparing the average expenses on weekdays and weekends.

Formula: Sample T-Test

$$\bar{x} = \frac{\sum x_i}{n}$$

R Program:

```
day <- c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday")

expense <- c(100, 150, 120, 180, 130, 300, 350)

data <- data.frame(day, expense)

print(data)

weekday <- data$expense[1:5] weekend <- data$expense[6:7]

average <- c(mean(weekday), mean(weekend))

result <- data.frame( Type = c("Weekdays", "Weekends"), Average_Expense = average )

print(result)

barplot(result$Average_Expense, names.arg = result$Type,

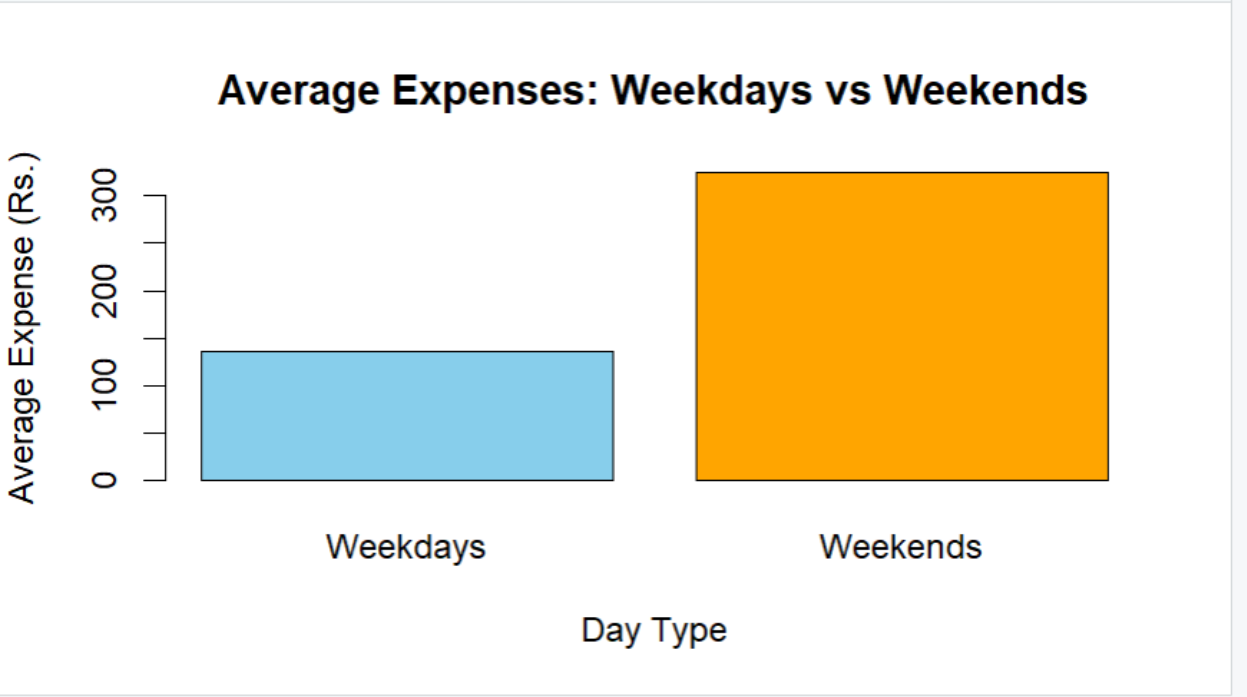
 col = c("skyblue", "orange"),

 main = "Average Expenses: Weekdays vs Weekends",

xlab = "Day Type",

 ylab = "Average Expense (Rs.)")
```

Output:



Question 4: A student records daily expenses for 10 days. Create a dataset and display a line graph to analyze daily spending trends and identify the day with the highest expense.

Formula:

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

R Program:

```
day <- 1:10

expense <- c(100, 150, 120, 200, 180, 250, 130, 300, 220, 350)
```

```
data <- data.frame(day, expense)

print(data)

hist(expense, main = "Daily Expense Analysis",

 xlab = "Expense Amount (Rs.)",

 ylab = "Frequency",

 col = "skyblue",

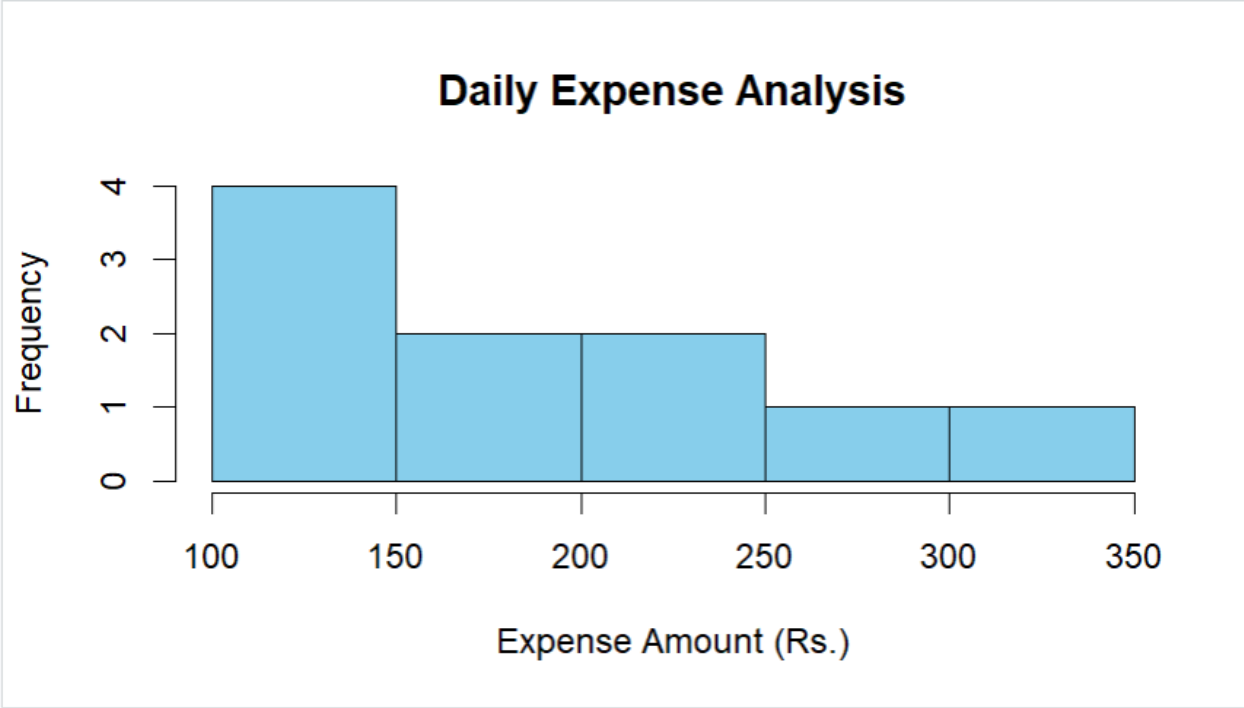
 border = "black")

highest <- which.max(expense)

cat("Day with highest expense:", day[highest], "\n")

 cat("Highest expense: Rs.", expense[highest], "\n")
```

Output::



Question 5: Create a dataset containing expenses for different categories over six months. Display a stacked bar chart to show how each category contributes to the total monthly expenses.

Formula:

F =MSB/MSw

R Program:

```
month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun")

Food <- c(2000, 2200, 2100, 2500, 2300, 2400)

Transport <- c(800, 900, 850, 1000, 950, 1100)

 Shopping <- c(1500, 1200, 1800, 1600, 2000, 1700)

Education <- c(2500, 2600, 2400, 2800, 2700, 3000)

data <- data.frame(month, Food, Transport, Shopping, Education)

print(data)

expenses <- as.matrix(data[, 2:5])

matplot(expenses, type = "p",

 pch = 19,

 col = c("red", "blue", "green", "orange"),

xaxt = "n",

main = "Monthly Expenses by Category",

 xlab = "Month",

ylab = "Expense Amount (Rs.)")
```

```
axis(1, at = 1:6, labels = month)

legend("topleft",

 legend = c("Food", "Transport", "Shopping", "Education"),

 col = c("red", "blue", "green", "orange"),

pch = 19)
```

Output:

